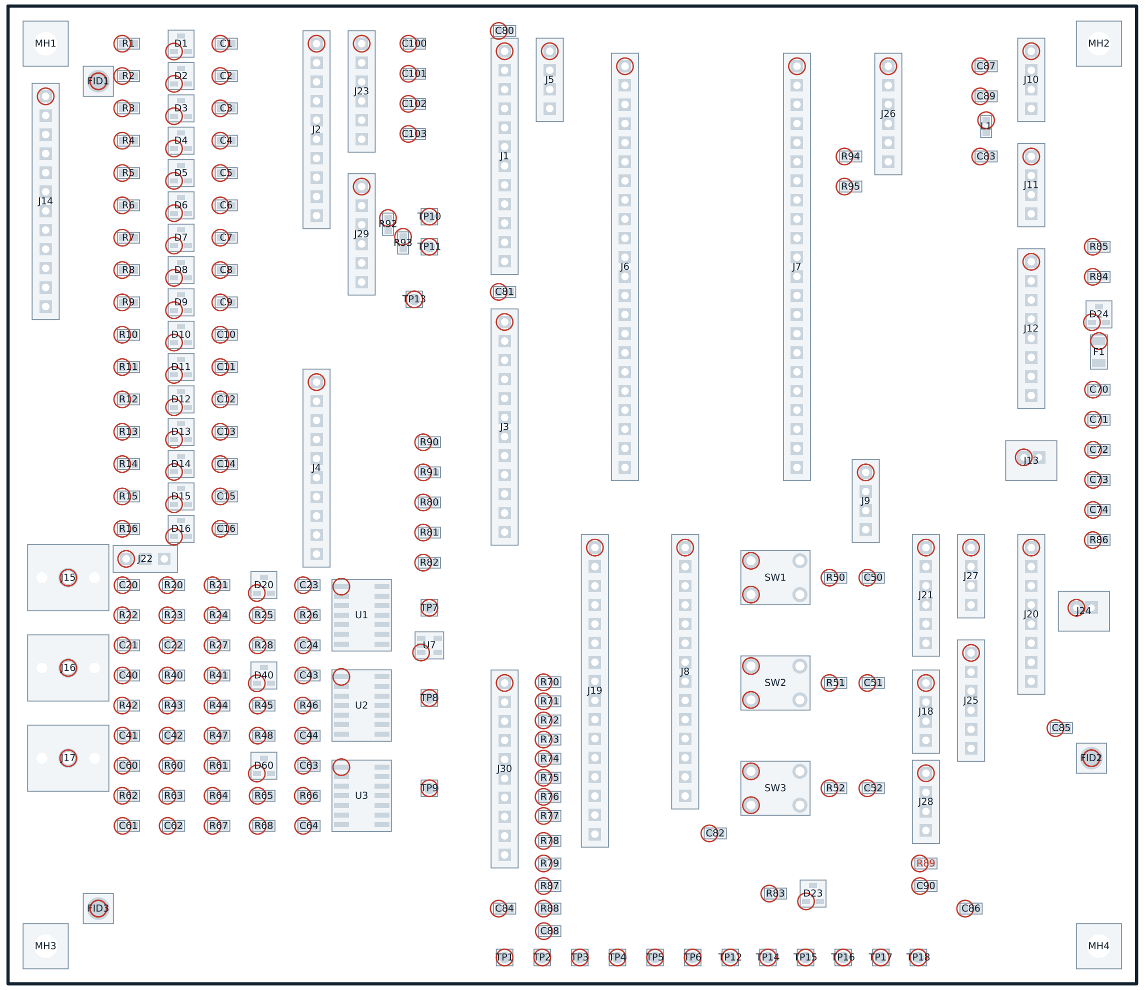

ASSEMBLY NOTES

SMT on the top side only. All through-hole parts on the top side.

Purchased modules do NOT plug directly into these sockets. They mount on the printed module plate MP-01 above the carrier and connect with keyed 2.54 mm ribbon jumpers per ICD-EEG-006. The one exception is the ESP32-S3-DevKitC-1, which is inserted directly into J6 and J7.

R90 and R91 are the single star points. Fit exactly one of each. Never bridge either with a wire link or a solder blob.

R92 and R93 are fitted by default. Remove both if ADS1299 module #2 regulates its own analogue rails; see ICD-EEG-006 section 4.

R89 is DO NOT POPULATE by default. Fit only if the boom preamp module does not supply its own electret bias.

Pin 1 of every socket strip is the square pad and is marked on the silkscreen.

PIN 1 IS RINGED IN RED ON EVERY POLARISED PART.

DO NOT POPULATE: R89

PROCESS: SMT reflow on the top side, then through-hole inserted from the top and soldered from the bottom. All parts are on the top side; the bottom side carries copper and legend only. The solder-side view is ASM 2 of 2 and it is MIRRORED.

GPIO AND MODULE NOTES

J6 and J7 are ESP32-S3-DevKitC-1 header POSITIONS, not GPIO numbers.

GPIO35, 36 and 37 carry the octal PSRAM on the -N16R8 variant and are NOT CONNECTED on the carrier (J7 positions 11, 12, 13). GPIO45 (J7 position 15) is the VDD_SPI strapping pin and is also left open.

Row spacing J6 to J7 is 22.86 mm (0.900 in), the DevKitC-1 header dimension.

The microSD interface is one-bit SDMMC. 70 kB/s is needed and about 2 MB/s is available, so the three data lines released by dropping four-bit mode are spent on the contact-light shift register instead.

SAFETY NOTES

Every conductor that can reach a person passes through a 47 kOhm 0.1 % series resistor (R1-R16) and a BAV99 clamp to AVDD/AVSS before it reaches a module.

AGND_REF is the analogue 0 V mid-rail. AVDD = +2.5 V and AVSS = -2.5 V are generated on ADS1299 module #1 and brought onto the carrier at J2 pins 11-14.

AGND_REF joins DGND at R90 ONLY. HARN_SHIELD joins DGND at R91 ONLY.

No copper crosses the isolation barrier. The ADuM4160 module at J10 is the only path between the host and the battery-powered side. Keep 8.0 mm clear of the J10 module outline on both layers -- the strip x >= 141 mm, y = 2 to 22 mm.

The charge input (J24) is a separate, charge-only USB-C receptacle. A session cannot start while VBUS is present (VBUS_DET) and the charger enable is held off by CHG_CE while a session is active. The helmet is never worn while charging.

EEG-CAR-01 -- assembly drawing, top side

Reference designators and orientation. Scale 1:1 on A3. Component side, as seen from above.

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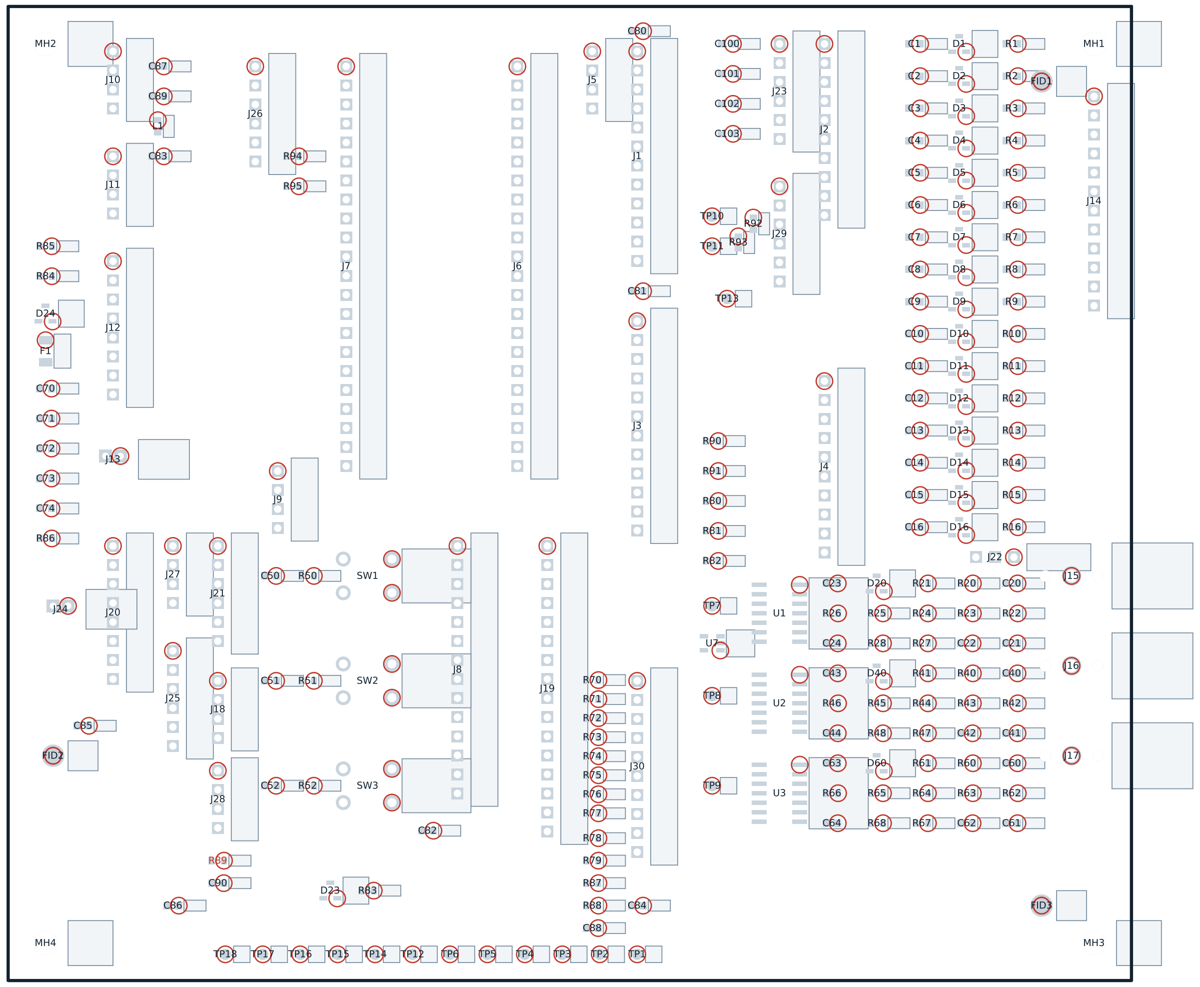

ASSEMBLY NOTES

SMT on the top side only. All through-hole parts on the top side.

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R89 is DO NOT POPULATE by default. Fit only if the boom preamp module does not supply its own electret bias.

Pin 1 of every socket strip is the square pad and is marked on the silkscreen.

PIN 1 IS RINGED IN RED ON EVERY POLARISED PART.

DO NOT POPULATE: R89

THIS SHEET IS MIRRORED. It is the view from UNDER the board, which is how the through-hole joints are soldered (ASM-EEG-007 section 2.7). Left and right are reversed against the top sheet. NO PART IS FITTED ON THIS SIDE -- the designators are shown so that a joint can be found from underneath.

GPIO AND MODULE NOTES

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EEG-CAR-01 -- assembly drawing, bottom side

Reference designators and orientation. Scale 1:1 on A3. MIRRORED -- drawn as seen from underneath, the soldering view.

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Rev B   2026-09-02   Sheet ASM 2 of 2